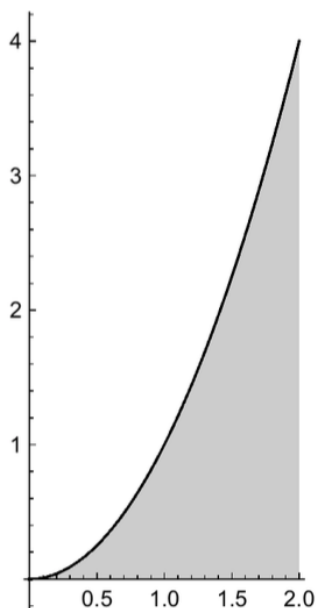


Problem 3

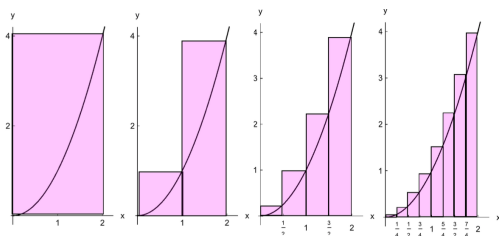
Problem 1 The graph of the function $f(x) = x^2$ is given in the figure.



Will a right Riemann sum approximation, for any value of n , be an underestimate or overestimate?

Explanation. Since the function f is positive and increasing on this interval, the right Riemann sum will always give an overestimate of the actual area.

Approximate the shaded area using a right Riemann sum with $n = 1, 2, 4$, and 8 rectangles, as illustrated in the figure below.



Problem 3

- (a) Approximate the shaded area using a right Riemann sum with $n = 1$ rectangles.

Explanation. $\Delta x = \frac{2}{1} = 2$

$$x_1^* = x_1 = 2$$

$$A \approx f(2)\Delta x = 4(2) = 8$$

- (b) Approximate the shaded area using a right Riemann sum with $n = 2$ rectangles.

Explanation. $\Delta x = \frac{2}{2} = 1$

$$x_k^* = x_k = k\Delta x = k(1), \text{ for } k = 1, 2$$

$$A \approx \sum_{k=1}^2 f(x_k^*)\Delta x = f(1)\Delta x + f(2)\Delta x = 1(1) + 4(1) = 5$$

- (c) Approximate the shaded area using a right Riemann sum with $n = 4$ rectangles.

Explanation. $\Delta x = \frac{2}{4} = \frac{1}{2}$

$$x_k^* = x_k = k\Delta x = k\left(\frac{1}{2}\right) = \frac{k}{2}, \text{ for } k = 1, 2, 3, 4$$

$$A \approx \sum_{k=1}^4 f(x_k^*)\Delta x = f\left(\frac{1}{2}\right)\Delta x + f(1)\Delta x + f\left(\frac{3}{2}\right)\Delta x + f(2)\Delta x = \Delta x \left(f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{3}{2}\right) + f(2) \right)$$

$$A \approx \frac{1}{2} \left(\frac{1}{4} + 1 + \frac{9}{4} + 4 \right) = \frac{5}{2} + 5 = \frac{15}{2}$$

- (d) Approximate the shaded area using a right Riemann sum with $n = 8$ rectangles.

Explanation. $\Delta x = \frac{2}{8} = \frac{1}{4}$

$$x_k^* = x_k = k\Delta x = k\left(\frac{1}{4}\right) = \frac{k}{4}, \text{ for } k = 1, \dots, 8$$

$$A \approx \sum_{k=1}^8 f(x_k^*)\Delta x = \Delta x \cdot \sum_{k=1}^8 f(x_k^*) = \frac{1}{4} \cdot \sum_{k=1}^8 f\left(\frac{k}{4}\right) = \frac{1}{4} \cdot \sum_{k=1}^8 \left(\frac{k}{4}\right)^2 =$$

$$\frac{1}{4} \cdot \sum_{k=1}^8 \frac{k^2}{16} = \frac{1}{64} \cdot \sum_{k=1}^8 k^2$$

Recall: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ Therefore

$$A \approx \frac{1}{64} \cdot \sum_{k=1}^8 k^2 = \frac{1}{64} \cdot \frac{8(9)(17)}{6} = \frac{1}{8} \cdot \frac{3(17)}{2} = \frac{51}{16}$$